Challenge Context
Good Nature Agro is a Zambian social enterprise that works with thousands of smallholder farmers across rural Zambia. GNA supports farmers through a structured input-loan and crop buyback model designed to increase productivity, income stability, and market access. Under this model, farmers receive:

• Input package loans and access to various asset loans
• High-touch Extension services and planting guidance throughout the growing season
• Crop buyback agreements at harvest where the farmers can repay their loans in kind (through crop) or cash and GNA purchases the net quantity of the farmers crop at competitive prices.

For GNA, accurately estimating expected yield per farmer is critical for:
- Improving extension targeting and support allocation, in real time and season over season
- Identifying farmer segments at risk of underperformance and how we can help to improve yields
- Optimizing input package design for maximizing yields and long-term soil health
- Forecasting total crop procurement volumes, directly informing logistics, processing, and sales
- Managing working capital and loan exposure
- Understanding true yields vs how much farmers sell on to GNA, helping to answer how much is held for consumption, recycling seed, or selling to other buyers
- Measuring the success of our input packages versus competitors'.

The challenge for students is to use real operational data to generate insights and predictive models that can support better decision-making and risk management.

1. Predictive Objective
Develop a model to predict expected farmer yield (or crop buyback potential) using the available data from the provided datasets. Evaluate model performance and identify the most significant predictors.

- total_weight = target variable
    - ways to predict total_weight
    -

Relevant fields for buyback predictions (need to correctly formulate the cash being generated from crops sold, how many crops are kept, etc.)
- Target - crop yield values and loan amounts
    - amount of crops (money they're worth) / land = yield/land
    - yield/land to amount of help/money and this would determine how easily they can pay back.
    - buyback_details.csv
        - grade_a_cash_value
        - grade_b_cash_value

        - grade_c_cash_value
        - grade_a_weight
        - grade_b_weight
        - grade_c_weight
    - loan_details.csv
        - initial_down_payment_value
        - package_cash_repayment
        - package_in_kind_repayment
    - use the buyback details and loan_details to see repayment possibilities
- Dependent variables
    - farmer_details.csv

2. Input Effectiveness
Which components of the loan package are most strongly associated with improved yield outcomes? Are there nonlinear or interaction effects?

- Feature Importance (For the loan package variables only)
- Check for residuals after running linear model, a scatter plot would look like a curve for nonlinear relationships
- For interaction effects, create faceted plots or interaction plots

3. Risk Identification
Can early-season data be used to identify farmers at high risk of underperformance? Can farmers be segmented by risk profile and can we flag high-risk farmers early in the season.
- If we create different risk values based on the buyback predictions (use predicted Y/N values or try to create risk clusters?)
- early season data
    - planting_survey.csv (all fields look relevant)

What's happening
- GNA is loaning money to farmers and wants to observe/predict which farmers are likely to be able to pay back their loans through crop or cash.

Parties involved
GNA (Good Nature Agro) - the company/organization we are researching for
Farmers
- farmer 1 - have land and farming experience
    - needs a loan
        - loan - typically, people won't give you money unless you prove the ability to pay it back or have something to put as collateral.
        - Collateral = promise of giving future crops and or sales from crops back
    - GNA - gives loan/aid/package (money, seeds, fertilizer, etc.)

        - for payment, GNA accepts the harvested crops or promises to purchase the crops at a set price based on quality.
    - **GNA is to figure out the likelihood that farmers will be able to repay the loans/value of the loans.**
        - Predictive model - tells us the amount of crops the farmer can grow and use that data to see whether or not they can likely pay the loan.
        - input effectiveness - if certain elements of a package prove to be more successful in helping farmers increase yield, it makes sense to increase those elements going forward.
        - risk identification - can we cluster different groups of farmers with different risk levels to see whether or not a loan is a bad idea or not or what groups require different actions or packages to better succeed.
- farmer 2

Questions to ask for office hours
- Is all the data for the same year?
    - are we able to look at year over year changes
    - can we assume that weather conditions and the like were stable across regions.
    - are there specific competitors they would like us to look at?
- Can a single farm have multiple farmers?
    - are we to assume this is recent data or does it belong to a single year

Information on the farmer and load details aggregated by crop class.

Planting survey on farmer id and crop crass - we have distinguished between the crop class in the planting survey.